Adding PostgreSQL Dependency to SpringBoot

1. Why do we need PostgreSQL dependency?

- Our SpringBoot application is a **JAVA application**
- **To communicate** with a PostgreSQL database, java needs a **PostgreSQL driver.**
- The PostgreSQL driver allows the JAVA application to connect and interact with PostgreSQL
- It is a **JDBCDriver**

2. Get the Dependency from Spring Intializr

- Go to **Spring Initializr** start.spring.io
- Select
  - **Maven** if your project uses maven
  - **Gradle** if your project uses gradle
- Go to **Add Dependencies**
- Search for **PostgreSQL**
- Select **PostgreSQL Driver**

3. Add dependency to pom.xml

- After selecting the **PostgreSQL Driver,**
- Now **Spring Initializr** shows the corresponding pom.xml
- Copy the PostgreSQL dependency
- Open your SpringBoot project's **pom.xml**
- Paste the dependency inside the <dependencies> section

Conceptually:

```
Spring Boot Application
        ↓
PostgreSQL Driver dependency
        ↓
Java application can communicate
with PostgreSQL
```

4. Reload Maven

- After modifying **pom.xml**
- IntelliJ may show a **Load Maven Changes/ Reload Maven icon**
- Click it to **reload** the **Maven Project**
- If the icon doesn’t appear
  - Open the **Maven tool** window
  - **Manually reload** the Maven Project

5. Verify the Dependency

- After Maven finishes resolving the dependencies
  - Go to **External Libraries** in IntelliJ
  - Search for **PostgreSQL**
- we should see the **PostgreSQL** dependency there
- This confirms that Maven has successfully downloaded/resolved the PostgreSQL driver.

6. Important Interview Point

What is PostgreSQL Driver?

- The PostgreSQL JDBC Driver is a dependency that allows a JAVA/SpringBoot application to communicate with a PostgreSQL database.

Remember the flow:

```
Spring Boot Application
        ↓
PostgreSQL Driver
        ↓
JDBC
        ↓
PostgreSQL Database
```

Key commands/concepts from this lecture

```
start.spring.io
       ↓
PostgreSQL Driver
       ↓
pom.xml
       ↓
Reload Maven
       ↓
External Libraries
       ↓
PostgreSQL dependency resolved
```